

■ HACKATHON SUBMISSION

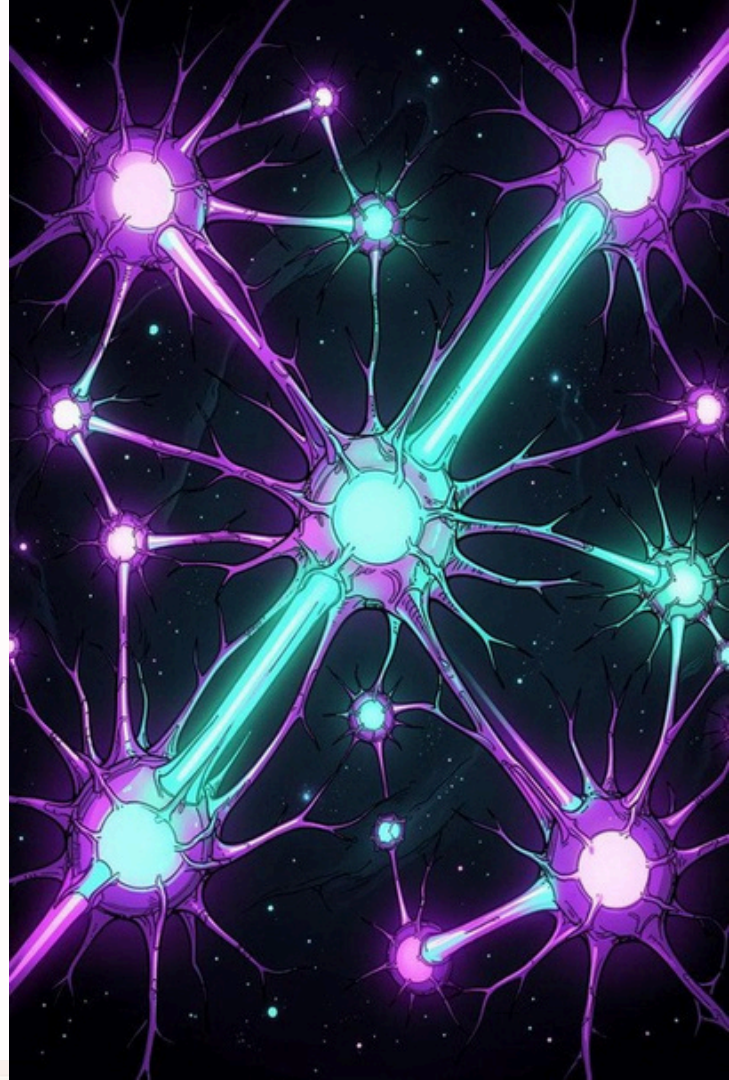
India Runs Data & AI
Challenge

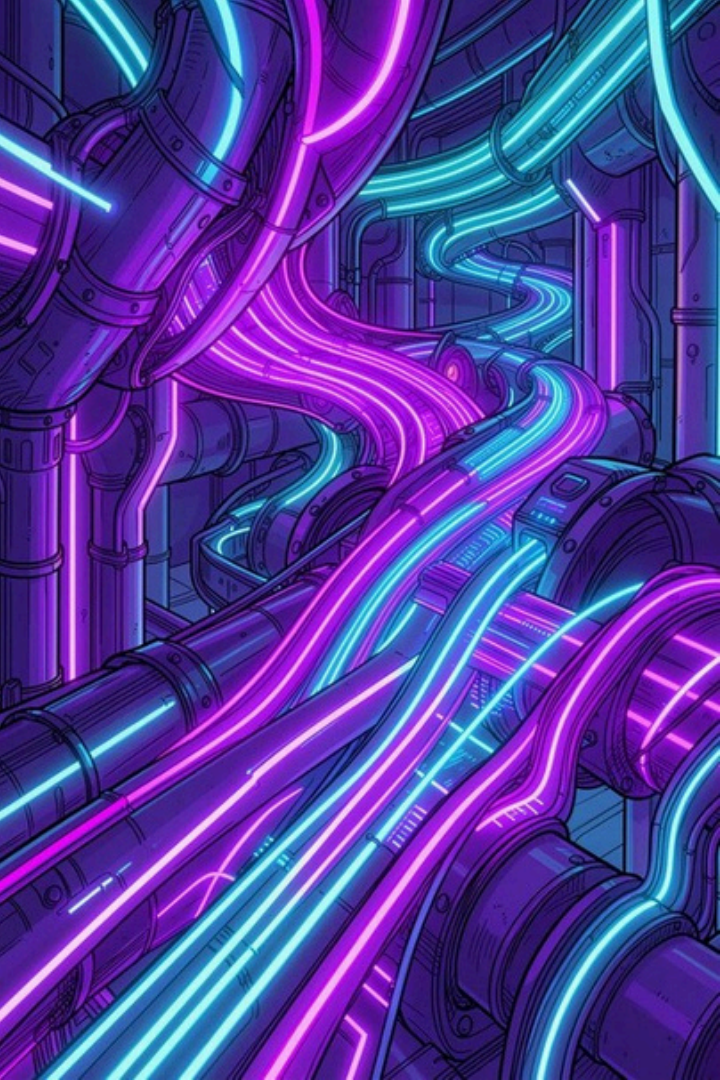
AI-Powered

Candidate Ranking
System

Intelligent, scalable, and transparent—ranking 100,000 profiles to surface the best Senior AI Engineer candidates. Prepared

by: [Duru shruthi]





PROBLEM STATEMENT

The Ranking Challenge

Given a large-scale candidate dataset, the goal is to intelligently rank and shortlist the most qualified profiles for a Senior AI Engineer role—with precision, speed, and a clean, reproducible output.

Senior AI



100,000 Profiles

Process a massive candidate dataset at scale



JD Matching

Align candidates to a Senior AI Engineer job description



Top 10 Output

Generate a ranked shortlist of the best-fit candidates



Valid submission.csv

Produce a clean, submission-ready CSV file

DATA

Dataset Overview

Eight rich data dimensions power the ranking model—each contributing unique signals to evaluate candidate quality.



Candidate Profiles

Core identity, location, and professional summary



Career History

Work experience, roles, and tenure data



Certifications

Professional and cloud certifications earned



GitHub Activity

Commits, repos, and open-source contributions



Skills

Technical competencies and domain expertise tags



Education

Degrees, institutions, and academic background



Recruiter Signals

Platform-specific recruiter engagement metrics

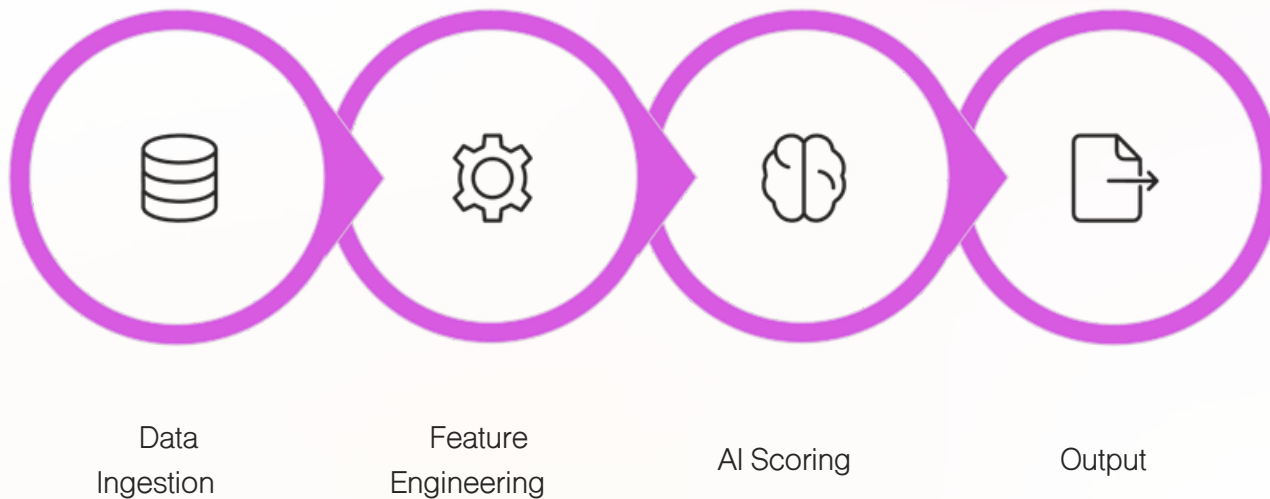


Recruiter Metrics

Response rates, profile views, and outreach data

Solution Architecture

A modular, end-to-end pipeline transforms raw candidate data into a ranked, submission-ready output — with every stage optimised for speed and accuracy.



Each stage is independently testable, enabling rapid iteration and transparent debugging during the hackathon sprint.



Technologies Used



Python

Core language for data processing, scoring logic, and pipeline orchestration



Pandas

High-performance data manipulation, filtering, and aggregation at scale



JSON

Structured data parsing for nested skill and profile fields



Google Colab

Cloud-based execution environment with GPU access for rapid prototyping



GitHub

Version control, collaboration, and public repository for reproducibility

ALGORITHM

Ranking Algorithm

Candidates are scored across **six weighted dimensions**, each reflecting a critical signal of Senior AI Engineer readiness. The final composite score determines the ranking order.

Experience (5- 9 Years)

Optimal seniority band for the role

Recruiter Sign als

Engagement and response metrics

Open-to-Work

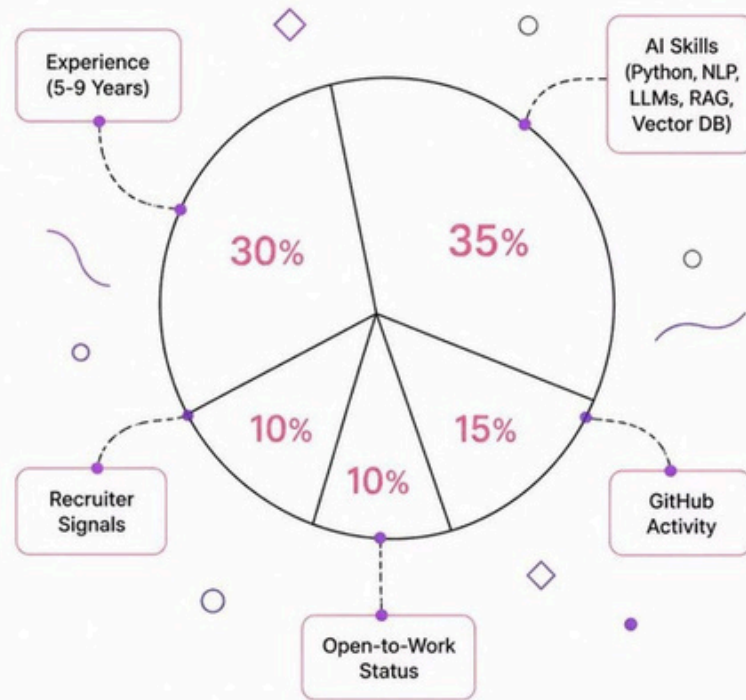
Immediate availability signal

AI S kills

Python, NLP, LLMs, RAG, Vector DB

GitHub Ac tivity

Code quality and open-source presence





RESULTS

Processing Results

100K

Full dataset ingested, cleaned, and scored

100

Candidates Ranked

Top 100 shortlist generated with composite scores

1

submission.csv

Valid, submission-ready output file produced

~0s

Processing Time

Optimised pipeline for fast, scalable execution

Advantages & Future Improvements

Current Advantages

Fast & Scalable
Handles 100K profiles efficiently

Interpretable Scoring
Interpretable, rule-based weights

AI-Focused
Optimised specifically for AI Engineer roles

Future Improvements

Embedding-Based Similarity
Vectorise skills and JD text

Resume-JD Similarity
Cosine similarity scoring

LLM-Based Ranking
Generative AI for nuanced evaluation

ML-Based Ranking
Learn weights from historical hires

GitHub Repository Structure

The project follows a clean, modular structure—making it easy to audit, extend, and reproduce. Every file has a clear purpose.

 Project Root

Organised file hierarchy for clarity

 submission.csv

Final ranked output — Top 100 candidates

 rank_candidates.py

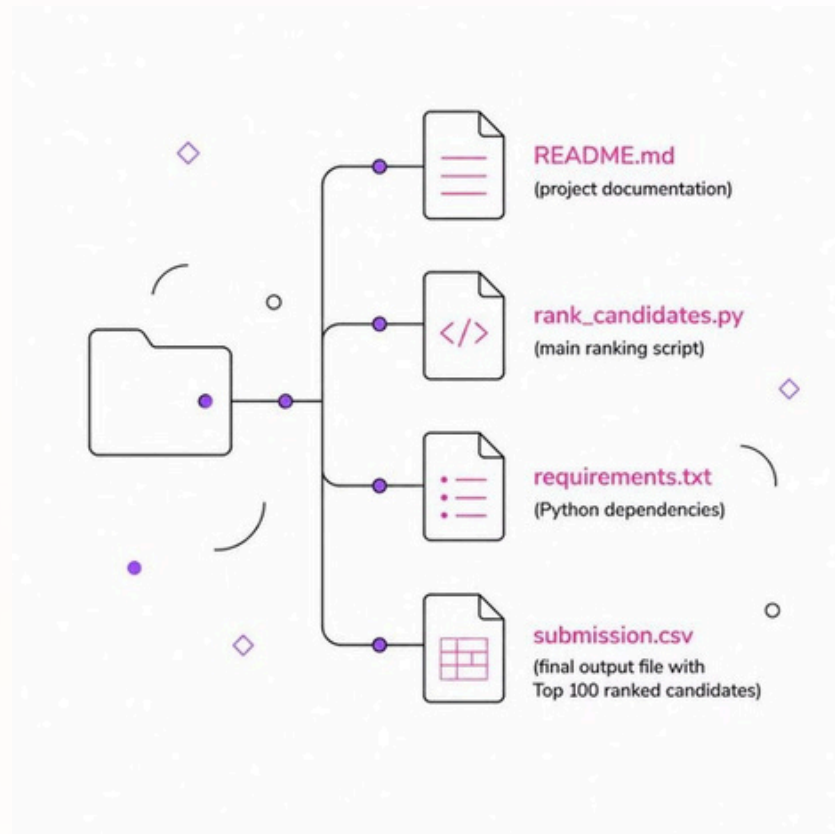
Core ranking logic and scoring pipeline

 README.md

Setup instructions, usage guide, and methodology

 requirements.txt

Reproducible environment with pinned dependencies



Thank You

*We built an intelligent, transparent, and scalable candidate ranking system —
purpose-built for the India Runs Data & AI Challenge.*

Questions?

Happy to walk through the architecture, scoring logic, or results in detail.

Contact

[Your Email]  *[Your LinkedIn]*  *[GitHub Repository Link]*



*India Runs Data & AI Challenge—AI-Powered Candidate Ranking
System  Hackathon Submission*